

“There is a **spring**, or **elastical power** in the air we live in.”

Robert Boyle, 1660

each other. In 1660, several members, including Boyle, founded the Royal Society, the world's first scientific body.

Experimental science

Using income from his estate to fund his endeavors, Boyle devoted himself to scientific research. With his assistant Robert Hooke (see pp.70–75), he devised an air pump, or vacuum chamber, which enabled him to study the physical properties of air. His experiments demonstrated that sound cannot travel through a vacuum, but that magnetic forces and light can. He also showed that combustion cannot take place in a vacuum.

In their most famous experiment, in 1662, Boyle and Hooke used mercury to alter the pressure on a fixed body of air and discovered that when the pressure of a gas increases, its volume decreases proportionally. This was the first universal law governing the behavior of gas and became known as Boyle's law.

Boyle successfully distinguished chemistry from the mystical practice of alchemy and established chemistry as a science in its own right. He rejected the old Aristotelian elements of air, fire, earth, and water and provided the first modern definition of an element as a substance that cannot be broken down into simpler constituents. He also argued that air was composed of tiny, moving “corpuscles” of matter, which paved the way for the development of modern kinetic theory. Like Galileo, Boyle believed that the natural world is governed by mathematics and wanted science to be a process of empirical investigation. He meticulously recorded every stage of his investigations and in doing so became a pioneer in modern experimental methods.

Boyle's work was at the heart of an intellectual movement that saw the development of new theories about the natural world. An innovative thinker who led the charge in experimental science, he achieved both national and international renown during his lifetime.

JOSEPH LOUIS GAY-LUSSAC

French chemist and physicist Joseph Louis Gay-Lussac was an early investigator into the behavior of gases; he established several laws that describe their behavior.



An accomplished scholar, Gay-Lussac (1778–1850) specialized in the behavior of gases. In 1802, he provided evidence to prove Charles's law, which said that gases expand by the same amount with the same rise in temperature. This was the first gas law since Boyle's law. He also established Gay-Lussac's law, which states that the pressure of a given mass of gas varies directly with the absolute temperature of the gas, when its volume is kept constant. Gay-Lussac's many investigations involved two ascents in a hydrogen balloon, one of which was a solo flight that reached a height of 23,000 ft (7,016 m).